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EUROPEAN UNION AVIATION SAFETY AGENCY
SAFETY INFORMATION BULLETIN • OPERATIONS & AIRPORTS

EASA SIB No.: 2024-09
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Subject: Mitigation of Ramp Collision Risks and Ground Vehicle Speed Management on Aprons

European Central Repository (ECR) Occurrence Analysis • Aerodrome Ground Operations Safety

REF. PUBLICATIONS & REGULATORY FRAMEWORK:

- **Regulation (EU) No 139/2014** (Aerodromes Regulation):
  – Annex III (Part-ADR.OR): Points ADR.OR.D.017 (Personnel Training) and ADR.OR.D.027 (Safety Reporting).
  – Annex IV (Part-ADR.OPS): Points ADR.OPS.B.025 (Operation of Vehicles), ADR.OPS.B.030 (Surface Movement), and ADR.OPS.B.070 (Works Safety).
- **Regulation (EU) No 376/2014** on the reporting, analysis, and follow-up of occurrences in civil aviation.
- **ICAO Doc 9137** (Airport Services Manual), Part 8 — Airport Operational Services.
- **European Plan for Aviation Safety (EPAS)** — Priority Ground Safety / Apron Operations.

TARGET APPLICABILITY:

National Competent Authorities (NCAs / CAAs / DSAC) • Aerodrome Operators certified under Regulation (EU) No 139/2014 • Ground Handling Service Providers (GHSPs) • Apron Management Service (AMS) units and Air Traffic Services (ATS) providers.

1. DESCRIPTION & OPERATIONAL CONTEXT

Analysis of aviation safety data reported to the **European Central Repository (ECR)** under Regulation (EU) No 376/2014 indicates a recurring, persistent trend of ground collisions and high-risk near-miss occurrences on aerodrome aprons across European airports. A substantial proportion of these occurrences involve **Ground Support Equipment (GSE)** and ground service vehicles colliding with parked or taxiing aircraft, passenger boarding bridges, mobile fueling tankers, and other servicing equipment.

Comprehensive contributing factor analyses conducted by EASA demonstrate that:

- **Excessive Driving Speed:** Ground vehicles frequently operate well above authorized ramp speed limits, severely extending braking distances on wet, contaminated, or rubberized pavement surfaces.
- **Turnaround Scheduling Pressures:** Tight 25- to 35-minute commercial aircraft turnarounds incentivize ground handling drivers to take unauthorized shortcuts through designated Equipment Clearance Areas (ECA).
- **Restricted Visual Scan Patterns:** High driver workload, blind spots around widebody fuselages and engine nacelles, and degraded visibility during night or winter operations represent primary causal precursors to aircraft structural impact.

2. SAFETY CONCERN

High-energy vehicle impacts in the congested ramp environment introduce immediate, critical threats to aviation safety and airworthiness:

- **Unreported Aircraft Structural Damage:** High-speed GSE impacts with aircraft fuselages, composite radomes, engine cowlings, or primary flight control surfaces (flaps, ailerons) have on multiple occasions gone undetected or deliberately unreported by ground handling staff fearful of punitive repercussions. As a consequence, commercial aircraft have departed with compromised structural integrity, creating severe in-flight decompression or controllability hazards.
- **Apron Incursions into Active Taxiways:** Ground vehicles traveling at excessive speeds have repeatedly failed to stop prior to entering active aircraft taxi lanes or taxiways, compromising mandatory separation distances with taxiing aircraft under ATS guidance.
- **Personnel Hazards & Ramp Injuries:** Elevated vehicle speeds inside congested Equipment Clearance Areas (ECA) and Equipment Staging Areas (ESA) correlate directly with severe ground personnel injuries and fatalities during pushback, baggage handling, and fueling operations.

OPERATIONAL WARNING: HIDDEN COMPOSITE DAMAGE

Modern aircraft utilizing carbon-fiber reinforced polymer (CFRP) fuselages (such as the Airbus A350 and Boeing 787) can sustain severe internal delamination from low-speed ground vehicle contact without visible exterior scratches. Any vehicle contact, regardless of speed, must trigger immediate structural inspection before flight release.

3. REGULATORY BASELINE (PART-ADR.OPS.B.025)

Pursuant to point **ADR.OPS.B.025 (Operation of vehicles)** of Regulation (EU) No 139/2014, the aerodrome operator must establish, verify, and enforce comprehensive rules governing the movement and speed of all vehicles operating on the movement area. Mandatory requirements include:

1. **Apron Driving Authorisation (ADA) Scheme:** All airside drivers must hold a valid ADA issued following certified theoretical instruction, practical airside driving exams, and regular refresher training in accordance with Point ADR.OR.D.017.
2. **Vehicle Roadworthiness & Safety Checks:** The operator must enforce scheduled maintenance inspections, mandatory brake efficiency testing, and lighting checks on all tenant GSE authorized to enter aircraft movement areas.
3. **Formal Enforcement & Sanctions:** The Aerodrome Manual must contain a clear points-based penalty system, including temporary driving suspensions and permanent permit revocations for speeding or dangerous maneuvering.

4. RECOMMENDATIONS

EASA recommends that Aerodrome Operators, in formal coordination with Ground Handling Service Providers (GHSPs) and National Competent Authorities (e.g., DSAC), immediately implement the following safety actions:

A. Infrastructure Design, Speed Limits & Zoning

1. **Differentiated Speed Regimes:** Establish and enforce clear, unambiguous speed limits tailored to specific apron zones:
   - *General Apron Roadways:* Maximum speed typically not exceeding **25–30 km/h**.
   - *Aircraft Stand / Servicing Footprint:* Strictly limited to **5–10 km/h (walking pace)** within the immediate vicinity of parked aircraft.
   - *Baggage Make-up Facilities & Tunnels:* Capped at **10–15 km/h** with mandatory convex blind-corner safety mirrors.
2. **Visual Markings & Road Signage:** Audit movement area road markings, ensuring speed limit roundels, stop bars, yield markings, and pedestrian crossings conform strictly to CS-ADR-DSN visual standards.
3. **Telematics & Electronic Speed Limiting:** Promote the mandatory adoption of physical or electronic speed governors on motorized GSE, as well as GPS geofencing technology that automatically restricts maximum vehicle speed when crossing into active aircraft stand perimeters.

B. Training, Authorisation & "Just Culture" Enforcement

1. **Rigorous ADA Practical Assessments:** Update driver licensing syllabi (ADR.OR.D.017) to include mandatory practical assessments on emergency stopping distances of loaded tractor-trailer trains under wet conditions, and blind-spot geometry around widebody aircraft engines and wingtips.
2. **Radar & Telematics Speed Surveillance:** Deploy active speed-detection devices (e.g., portable radar speed guns, fixed roadside speed cameras, or automated fleet telematics) on major apron corridors to systematically identify non-compliant operators.
3. **Transparent Non-Punitive Reporting Culture:** Reiterate across all handling teams that low-speed vehicle contacts with aircraft (even superficial paint scuffs) must be reported immediately without fear of punitive action. This guarantees certified maintenance inspections take place prior to departure release.

> **JUST CULTURE PRINCIPLE (REGULATION (EU) NO 376/2014)**
> Frontline ground staff who report an occurrence or ground collision in accordance with published internal reporting procedures shall not be subject to any prejudice or punitive measures by their employer, except in cases of gross negligence or willful misconduct.

4. RECOMMENDATIONS (CONTINUED)

C. Stakeholder Coordination via the Ramp Safety Committee (RSC)

1. **Regular Committee Convening:** Convene regular monthly sessions of the local Airport Ramp Safety Committee (RSC) uniting the aerodrome operator, airline representatives, ground handlers, and air traffic control. The RSC must review radar surveillance metrics, fleet telematics exceedances, and ramp incident reports.
2. **Joint Safety Audits:** Establish joint safety surveillance audits between the aerodrome operator and tenant ground handling service providers to ensure third-party contractors adhere to vehicle speed rules published in the Aerodrome Manual.
3. **Collaborative Hazard Mitigation:** Coordinate immediate infrastructure modifications (e.g., relocation of staging areas, installation of physical barrier bollards, or improved apron floodlighting) when near-miss clustering is identified on specific aircraft stands.

5. STATUS & ADMINISTRATIVE PROVISIONS

Informative Nature of this Safety Bulletin:

- This Safety Information Bulletin (SIB) is informative in nature and non-mandatory. It does not constitute a formal European Union regulation, an Implementing Rule, or an EASA Certification Specification.
- At this time, the safety condition described does not warrant the issuance of an urgent Safety Directive (SD) under Regulation (EU) No 139/2014, Annex II, point ADR.AR.A.040.
- However, National Competent Authorities (NCAs) are strongly encouraged to verify the adoption of these recommended speed management controls during upcoming annual aerodrome oversight audits.

CONTACT & INQUIRIES

For technical questions or operational feedback regarding the safety recommendations contained in this SIB, please contact:

EASA Safety Intelligence & Performance Department
Executive Directorate • European Union Aviation Safety Agency
Postal Address: Konrad-Adenauer-Ufer 3, D-50668 Cologne, Germany
Functional Mailbox: **ADR@easa.europa.eu** / **safety.information@easa.europa.eu**

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